

Sao Aphisith Sithisack

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EDUCATION

University of Oklahoma

B.S. in Electrical & Computer Engineering (Honors) | Minor in Mathematics & Computer Science | GPA: 4.0

Norman, OK

Aug 2024 – Aug 2028

RELEVANT EXPERIENCE

Undergraduate Research Assistant

Jan 2026 – Present

Pulsed Laser Template Engineering (PLATEN)

Norman, OK

- Fabricate **high-aspect-ratio (HAR) micro- and nanostructures** via photolithography, reactive ion etching (RIE), and pulsed laser deposition (PLD).
- Optimize anisotropic plasma etch parameters (gas chemistry, RF power, pressure) to achieve vertical sidewalls and controlled feature depth.
- Support end-to-end process development including resist patterning, etch selectivity, and post-process inspection.

Electrical Systems Member

Jan 2026 – Present

Sooner Competitive Robotics

Norman, OK

- Designed power distribution architecture and CAD layouts for electrical component placement and wire routing.

Peer Learning Assistant

Aug 2025 – Present

Student Learning Center

Norman, OK

- Mentored 250+ students per semester in Physics I–II, Calculus I–III, and General Chemistry I using structured quantitative problem-solving methods.

Research Apprentice

Aug 2025 – Dec 2025

Light–Matter Interaction Research (Dr. Tischler)

Norman, OK

- Reviewed light–matter interactions in silicon carbide nanocrystals and gained exposure to optical spectroscopy systems.

Research Apprentice

Jan 2025 – May 2025

First Year Research Experience – Cloud Microphysics Group

Norman, OK

- Modeled atmospheric profiles using Python (NumPy, Pandas, Matplotlib) with numerical methods and statistical analysis.

PROJECTS

Electrical Engineering Planner

Apr 2026

Personal Project

GitHub

- Designed a structured project planner that curates domain-specific engineering curricula across VLSI, Hardware Engineering, Microelectronics, and Photonics.
- Integrated Python-based CMOS simulations (NumPy, Matplotlib) and industry EDA tools (Cadence Virtuoso, Magic VLSI, Vivado) to guide hands-on IC design projects from concept to implementation.

Job Application Agent

Mar 2026 – Present

Personal Project

GitHub

- Built an autonomous 7-phase Python pipeline that parses resumes, scores jobs across 6 weighted criteria, tailors application materials, and generates Excel trackers with 3 pluggable LLM backends (Claude API, Ollama, offline regex).

Climb – Hacklahoma Project

Feb 2026

Personal Project

GitHub

- Pomodoro productivity app built with Expo, React Native, and TypeScript featuring a reward/penalty system for app usage restriction.

ADDITIONAL EXPERIENCE

Alumni Coordinator / Board Member

Jul 2024 – Present

United World College Lao PDR National Committee

Hybrid

- Led fundraising exceeding \$2,500 and coordinated communication among alumni, scholars, and partner organizations.

Sponsorship & Volunteering Chair

Nov 2024 – Present

Association of Southeast Asian Nations at OU

Norman, OK

- Secured \$1,000+ in external funding through outreach to 20+ donors and organized cultural events.

TECHNICAL SKILLS

Python (NumPy, Pandas, Matplotlib) | Statistical Analysis | Data Visualization | AI-Assisted Data Analysis | LLM Integration (Claude API, Ollama) | Optical Spectroscopy | Spectrometers | Pulsed Laser Deposition (PLD) | Numerical Methods | MATLAB | LabVIEW | Photolithography | Anisotropic RIE | Cleanroom Processing | CAD Layout Design | LTSPICE | Java

INTERESTS

Photonics | Optoelectronics | Semiconductor Manufacturing | High-Aspect-Ratio Devices | Microelectronics | VLSI Design | Integrated Circuit Design | Hardware Engineering